PROBLEM CL7

Industrial Dryer: Velocities, Force, Power and Torque

Original statement · CL7.pdf

Axisymmetric | Angular momentum | Energy | Ideal gas

OFFICIAL PROBLEM STATEMENT

The axisymmetric industrial dryer of the figure in CL7.pdf is designed to provide a steady hot jet of radius R, mass flux G, and temperature T_s. Air enters the dryer from the ambient, at temperature and pressure T_a and p_a, through a radial inlet of surface A. Inside the dryer there exists an electric resistance that releases an amount of heat Q per unit time, as well as a fan of power W (work per unit time). Because of the presence of the fan, the air velocity at the exit has both azimuthal and axial components, v_θ and v_z, such that $v_\theta/v_z = \tan(\beta)$. In the analysis, assume that the fan and the inner walls of the dryer are thermally insulated, that the air behaves as an ideal gas with constant R_g and specific heat at constant pressure c_p, that gravity has a negligible effect on the fluid motion, and that the flow properties are uniform at the inlet and outlet sections. For given values of G, T_s, T_a, p_a, A, R, R_g, c_p and $\tan(\beta)$, determine:

- The velocity at the entrance section v_r as well as the two velocity components at the exit v_θ and v_z.
- The force acting on the dryer, indicating clearly its magnitude and direction.
- The total power required to run the dryer $Q + W$.
- The torque acting on the dryer.

Source: CL7.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.35, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady (on average) axisymmetric flow; ideal gas $p = \rho R_g T$, so $h = e + p/\rho = c_p T$; neglect $\bar{g}$; uniform properties at inlet and outlet.

- Inlet: air taken from the ambient, with $p = p_a, T = T_a, \rho_a = p_a/(R_g T_a)$ and radial velocity v_r pointing inwards (as in the official solution). Exit: free jet, $p = p_a, T = T_s$, $\rho_s = p_a/(R_g T_s)$, velocity $v_z \bar{e}_z + v_\theta \bar{e}_\theta$ uniform over the disc of radius R.
- G is the mass flow rate of the jet (kg/s), as used in the official solution ($G = \rho_a v_r A$). Fan and inner walls insulated; the resistance releases $\dot{Q}$; the fan delivers the power $\dot{W}$ to the air.
- Data: $[G,\ T_s,\ T_a,\ p_a,\ A,\ R,\ R_g,\ c_p,\ \tan\beta]$. Unknowns: $[v_r,\ v_z,\ v_\theta,\ \bar{F},\ \dot{Q}+\dot{W},\ \bar{M}]$.

Symbol	Meaning	SI unit
G	mass flow rate	kg/s
A, R	inlet area, jet radius	m^2, m
v_r, v_z, v_θ	inlet and exit velocities	m/s
R_g, c_p	gas constant, specific heat	J/(kg·K)
$\bar{F}$, $\bar{M}$	force and torque on the dryer	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the dryer; cylindrical basis $(\bar{e}_r, \bar{e}_\theta, \bar{e}_z)$ with $\bar{e}_z$ along the axis in the jet direction; origin on the axis. By axisymmetry $\int \bar{e}_r \, d\theta = \int \bar{e}_\theta \, d\theta = 0$ over a full turn.
- Surfaces: Σ_1 (radial inlet, area A, $\bar{n} = \bar{e}_r$, $\bar{v} = -v_r \bar{e}_r$), Σ_2 (exit disc, $\bar{n} = \bar{e}_z$), Σ_w (fixed casing and resistance, $\bar{v} = 0$) and Σ_m (fan blades, $\bar{v} = \bar{v}_c$).

Force and torque of the air (inside, and ambient outside) on the dryer, and the heat and work exchanged:

$$\boxed{\bar{F} = \int_{\Sigma_w, \Sigma_m} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w, \Sigma_m} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_a(-v_r)A = -\rho_a v_r A$
Σ_2	$\rho_s v_z \pi R^2$
$\Sigma_w,\ \Sigma_m$	0

$$G = \frac{p_a}{R_g T_a}\,v_r A = \frac{p_a}{R_g T_s}\,v_z \pi R^2 \implies \boxed{v_r = \frac{G R_g T_a}{p_a A}, \qquad v_z = \frac{G R_g T_s}{p_a \pi R^2}, \qquad v_\theta = v_z \tan\beta} \quad (1)$$

Linear momentum (Notes Eq. 3.35)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_a v_r^2 \int_{\Sigma_1}\bar{e}_r\,d\sigma = 0$ (axisymmetric)	0 ($p = p_a$)	0
Σ_2	$\rho_s v_z \int_{\Sigma_2}(v_z\bar{e}_z + v_\theta\bar{e}_\theta)\,d\sigma = \rho_s v_z^2 \pi R^2\,\bar{e}_z$	0 (free jet)	0
$\Sigma_w,\ \Sigma_m$	0	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}$. On Σ_2, $\int_{\Sigma_2} \bar{e}_\theta \, d\sigma = 0$, so the swirl gives no force.

$$\rho_s v_z^2 \pi R^2 \, \bar{e}_z = -\bar{F} \implies \boxed{\bar{F} = -\frac{p_a}{R_g T_s} \, v_z^2 \, \pi R^2 \, \bar{e}_z = -G \, v_z \, \bar{e}_z} \qquad (2)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \left(e + \tfrac{v^2}{2}\right) dV}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(e + \frac{v^2}{2}\right) [(\bar{v} - \bar{v}_c) \cdot \bar{n}] \, d\sigma = -\int_{\Sigma_c} p \, \bar{v} \cdot \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma + \underbrace{\int_{V_c} \rho \bar{g} \cdot \bar{v} \, dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q} \cdot \bar{n} \, d\sigma + \underbrace{\int_{V_c} Q_R \, dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(e + \frac{v^2}{2}\right) (\bar{v} \cdot \bar{n}) \, d\sigma$	$-\int p \, \bar{v} \cdot \bar{n} \, d\sigma$	$\int \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma$	$-\int \bar{q} \cdot \bar{n} \, d\sigma$
Σ_1	$-G \left(e_a + \frac{v_r^2}{2}\right)$	$+p_a v_r A$	0	0
Σ_2	$+G \left(e_s + \frac{v_\theta^2 + v_z^2}{2}\right)$	$-p_a v_z \pi R^2$	0	0
Σ_m	0	$-\int_{\Sigma_m} p \, \bar{v} \cdot \bar{n} \, d\sigma$	$\int_{\Sigma_m} \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma$	0 **(insulated)**
Σ_w	0	0	0	$-\int_{\Sigma_w} \bar{q} \cdot \bar{n} \, d\sigma$

Boxed terms: the work on Σ_m is the fan power $\dot{W}$; the heat through Σ_w (only the resistance, the walls are insulated) is $\dot{Q}$. With $p_a v_r A = G \, p_a/\rho_a$ and $p_a v_z \pi R^2 = G \, p_a/\rho_s$, $e + p/\rho = c_p T$:

$$\boxed{\dot{Q} + \dot{W} = G \left[\left(c_p T_s + \frac{v_\theta^2 + v_z^2}{2} \right) - \left(c_p T_a + \frac{v_r^2}{2} \right) \right]} \qquad (3)$$

Angular momentum about the origin, z component (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \bar{x} \wedge \bar{v} \, dV}_{0} + \int_{\Sigma_c} \rho \, (\bar{x} \wedge \bar{v}) [(\bar{v} - \bar{v}_c) \cdot \bar{n}] \, d\sigma = -\int_{\Sigma_c} \bar{x} \wedge (p - p_a) \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{x} \wedge (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})_z\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	pressure and viscous moments
Σ_1	0 ($\bar{x}\wedge(-v_r\bar{e}_r)=-zv_r\,\bar{e}_\theta$, no z component)	0
Σ_2	$\int_0^R \rho_s v_z\,(rv_\theta)\,2\pi r\,dr=\rho_s v_z v_\theta\,\frac{2\pi R^3}{3}$	0
$\Sigma_w,\ \Sigma_m$	0	$-\bar{M}\cdot\bar{e}_z$

On Σ_2, $\bar{x}=r\bar{e}_r+z_2\bar{e}_z$ and $\bar{x}\wedge(v_z\bar{e}_z+v_\theta\bar{e}_\theta)=-rv_z\bar{e}_\theta+rv_\theta\bar{e}_z-z_2v_\theta\bar{e}_r$; only $rv_\theta\bar{e}_z$ survives the integral. The antiderivative of r^2 is $r^3/3$, and Barrow's rule between 0 and R gives $R^3/3$.

$$\boxed{\bar{M}=-\rho_s v_z v_\theta\,\frac{2\pi R^3}{3}\,\bar{e}_z=-\frac{2}{3}\,R\,G\,v_\theta\,\bar{e}_z} \qquad (4)$$

Erratum in the official solution (slip in the boxed result)

CL7.pdf writes the momentum balance as $\frac{p_a}{R_gT_s}v_z^2\pi R^2\bar{e}_z=-\bar{F}$ but boxes $\bar{F}=-\frac{p_a}{R_gT_s}v_z^2\bar{e}_z$, without the area πR^2. That expression has units of pressure, not force; the correct result is (2).

PHASE 4

Interpretation, Limits and Dimensional Check

- The radial inflow gives no net momentum (axisymmetry), so the force is the reaction to the jet: magnitude Gv_z, direction $-\bar{e}_z$ (opposite to the jet).
- The swirl gives no force but gives a torque: the fan sets the air rotating, and the air reacts on the dryer with $\bar{M}$ opposite to the swirl. For $\beta=0$, $v_\theta=0$ and $\bar{M}=0$.
- (3): the electric and mechanical power go into heating the air from T_a to T_s and into its kinetic energy.
- Dimensions (SI): $[Gv_z]=\mathrm{N}$; $[RGv_\theta]=\mathrm{m}\cdot(\mathrm{kg/s})(\mathrm{m/s})=\mathrm{N}\cdot\mathrm{m}$; $[Gc_pT]=\mathrm{W}$.

- Numerical illustration (hypothetical data): $G = 0.5$ kg/s, $T_s = 350$ K, $T_a = 293$ K, $p_a = 101325$ Pa, $A = 0.05$ m^2, $R = 0.05$ m, $R_g = 287$ J/(kg K), $c_p = 1005$ J/(kg K), $\tan\beta = 0.5$ give $v_r = 8.2991$ m/s, $v_z = 63.112$ m/s, $v_\theta = 31.556$ m/s, $|\bar{F}| = 31.556$ N, $\dot{Q} + \dot{W} = 29870$ W and $|\bar{M}| = 0.52594$ N·m (a midpoint quadrature of the torque integral gives the same value).